

Assignment 9: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A09_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, zoo, and trend packages
 - Set your ggplot theme

```
#install.packages("zoo")  
#install.packages("trend")  
#install.packages("Kendall")
```

```
library(Kendall)
```

```
## Warning: package 'Kendall' was built under R version 4.5.3
```

```
library(here)
```

```
## here() starts at C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026
```

```
library(zoo)
```

```
## Warning: package 'zoo' was built under R version 4.5.3
```

```
##  
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':  
##  
##   as.Date, as.Date.numeric
```

```
library(trend)
```

```
## Warning: package 'trend' was built under R version 4.5.3
```

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(lubridate)
```

```
##  
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':  
##  
##   date, intersect, setdiff, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v forcats 1.0.0      v stringr 1.6.0  
## v ggplot2 4.0.0      v tibble 3.3.1  
## v purrr 1.1.0        v tidyr 1.3.2  
## v readr 2.1.5
```

```
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
```

```
ozone.2010 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"))
ozone.2011 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"))
ozone.2012 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"))
ozone.2013 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"))
ozone.2014 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"))
ozone.2015 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"))
ozone.2016 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"))
ozone.2017 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"))
ozone.2018 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"))
ozone.2019 <- read.csv(here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"))

GaringerOzzy <- bind_rows(ozone.2010, ozone.2011, ozone.2012, ozone.2013, ozone.2014, ozone.2015,
                          ozone.2016, ozone.2017, ozone.2018, ozone.2019)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3
```

```
GaringerOzzy$Date <- mdy(GaringerOzzy$Date)
```

```
# 4
```

```
GaringerOzzy.Clean <- GaringerOzzy %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE) %>%
  rename(Ozone.Concentrate = Daily.Max.8.hour.Ozone.Concentration)
```

```
# 5
```

```
Days <- data.frame(Date = seq(as.Date("2010-01-01"), as.Date("2019-12-31"), by="day"))
```

```
# 6
```

```
GaringerOzone <- left_join(Days, GaringerOzzy.Clean, by = "Date")
```

#Note to self: this left_join order keeps everything in the Days dataframe and matches rows from GaringerOzzy.Clean

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

#7

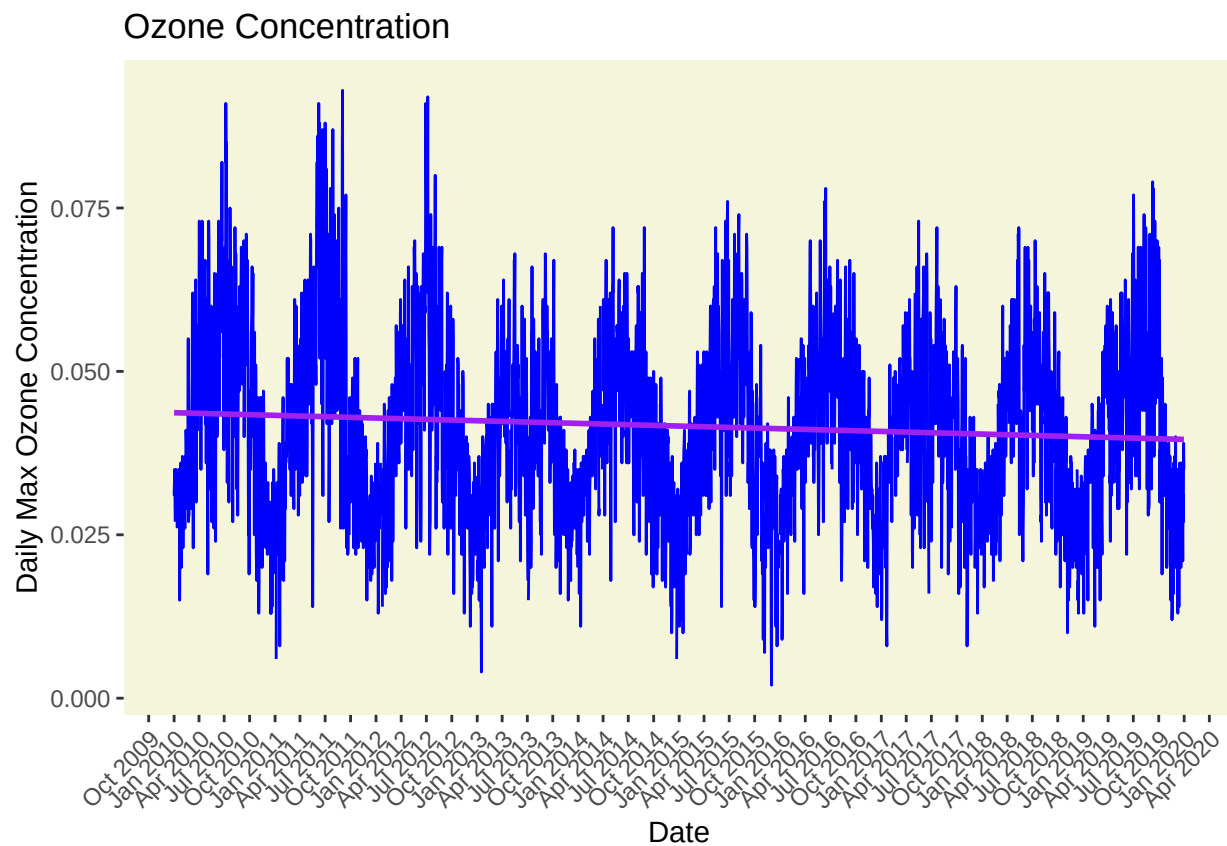
```
GaringerOzone.Plot <- GaringerOzone %>% ggplot(aes(x=Date, y=Ozone.Concentrate))+
  geom_line(color="blue")+
  scale_x_date(date_breaks = "3 months", date_labels = "%b %Y")+
  geom_smooth(color="purple",method = "lm", se=FALSE)+
  labs(
    title = "Ozone Concentration",
    x = "Date",
    y= "Daily Max Ozone Concentration"
  )+
  theme(panel.background = element_rect(fill = "beige"),
        panel.grid = element_blank(),
        axis.text.x = element_text(angle = 45, hjust = 1))

GaringerOzone.Plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```



Answer: The linear trend indicates a modest decrease in ozone concentration over time.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

#8

```
summary(GaringerOzone) #there are 63 NA values.
```

```
##      Date      Ozone.Concentrate DAILY_AQI_VALUE
## Min.   :2010-01-01 Min.   :0.00200 Min.   : 2.00
## 1st Qu.:2012-07-01 1st Qu.:0.03200 1st Qu.: 30.00
## Median :2014-12-31 Median :0.04100 Median : 38.00
## Mean   :2014-12-31 Mean   :0.04163 Mean   : 41.57
## 3rd Qu.:2017-07-01 3rd Qu.:0.05100 3rd Qu.: 47.00
## Max.   :2019-12-31 Max.   :0.09300 Max.   :169.00
##      NA's      :63      NA's      :63
```

#We can use a linear interpolation because the data follows a predictable trend and the total change in

```
GaringerOzone.Interpol <- #this is the daily data
  GaringerOzone %>%
  mutate(Ozone.Concentrate.Interpol = zoo::na.approx(Ozone.Concentrate)
  )
```

```
summary(GaringerOzone.Interpol$Ozone.Concentrate.Interpol)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

Answer: Linear interpolation is a solid option because the data follows a gradual trend, which makes it a practical option to fill in the gaps in data. Piecewise constant would create step-ladder jumps, and the spline interpolation would tend toward a curve shape that doesn't align with the pattern in the existing data.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

#9

```
GaringerOzone.monthly <- GaringerOzone.Interpol %>%
  mutate(the_month=month(Date), the_year=year(Date)) %>%
  group_by(the_year, the_month) %>%
  summarise(mean_ozone=mean(Ozone.Concentrate.Interpol))
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by the_year and the_month.
## i Output is grouped by the_year.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(the_year, the_month))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.
```

```
Graph.GaringerOzone.monthly <- GaringerOzone.monthly %>%
  mutate(graph_date = as.Date(paste(the_year, the_month, "1", sep = "-"))) #didn't work until i added the
```

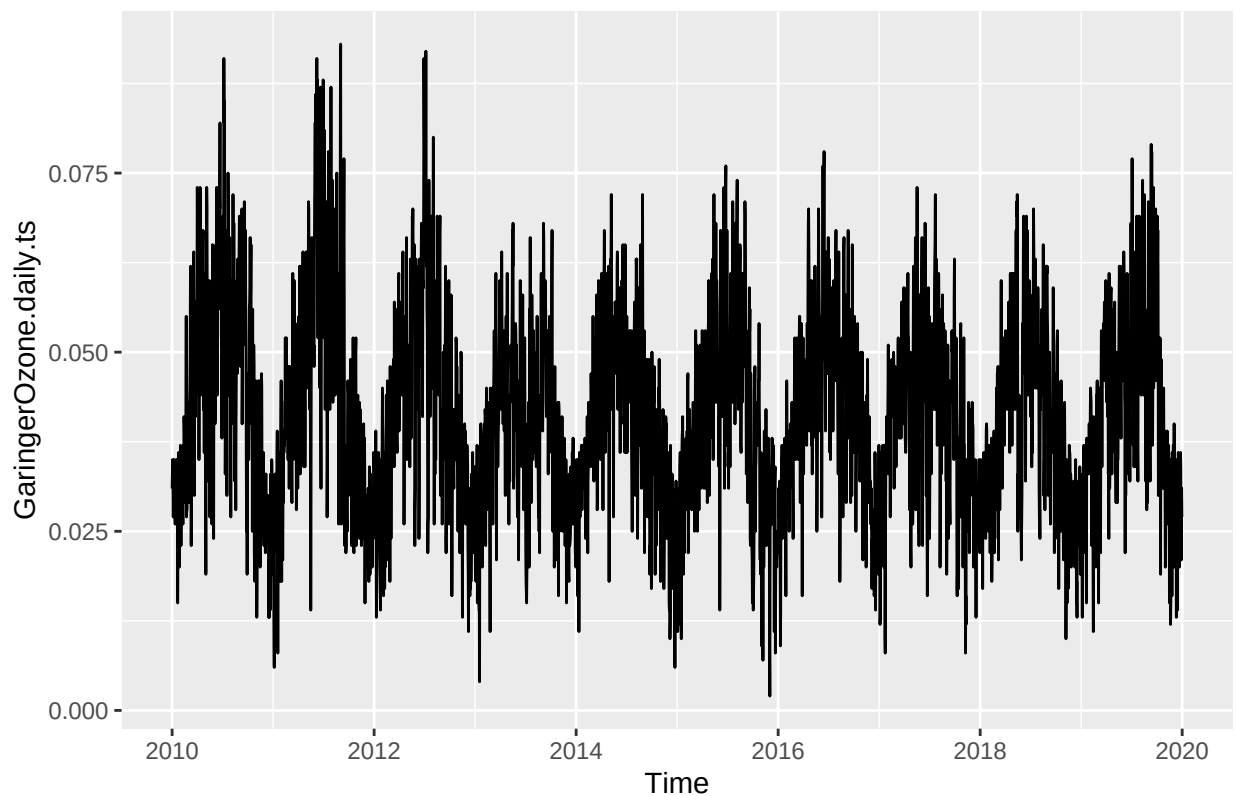
10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10
```

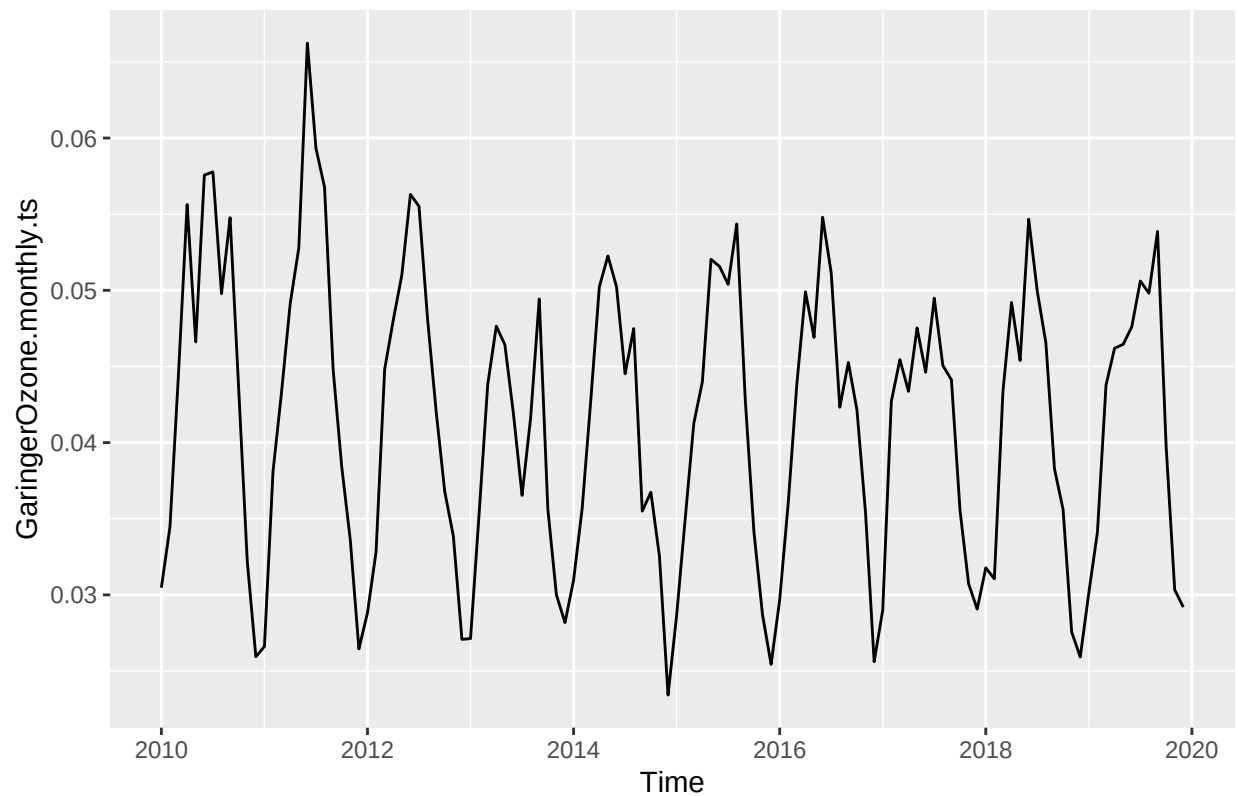
```
GaringerOzone.daily.ts <- ts(data = GaringerOzone.Interpol$Ozone.Concentrate.Interpol, start = c(2010, 1),
  GaringerOzone.monthly.ts <- ts(data = Graph.GaringerOzone.monthly$mean_ozone, start = c(2010, 1), end = c(2020, 12),
  #install.packages("forecast")
  library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.5.3
```

```
autoplot(GaringerOzone.daily.ts)
```



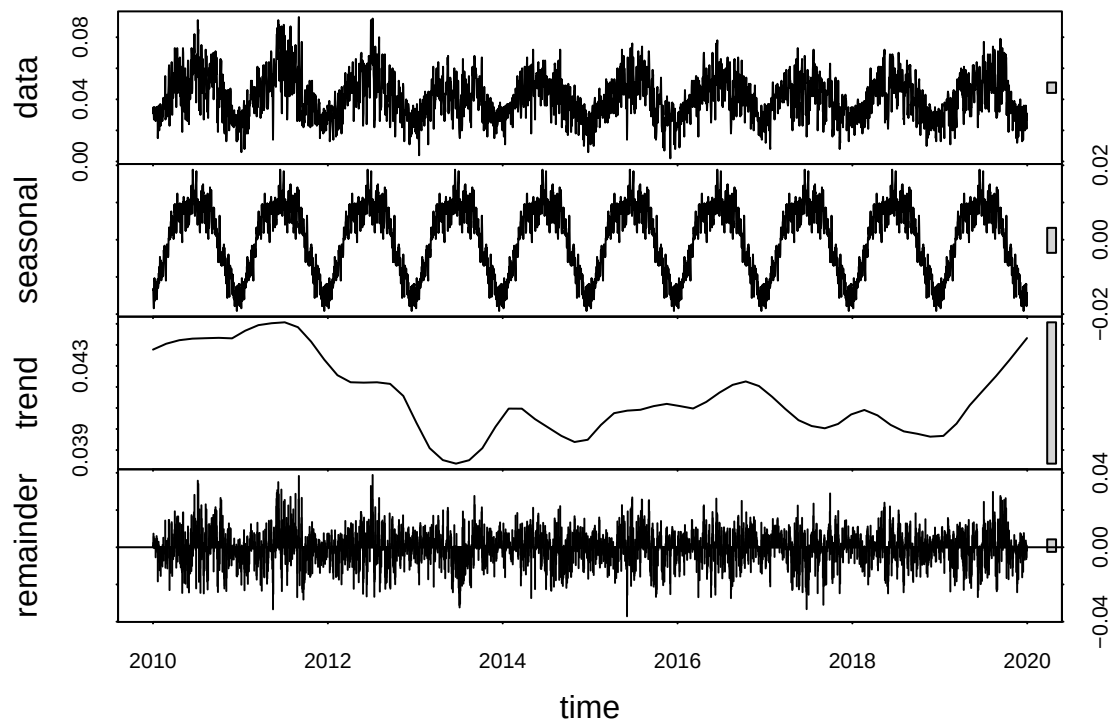
```
autoplot(GaringerOzone.monthly.ts)
```



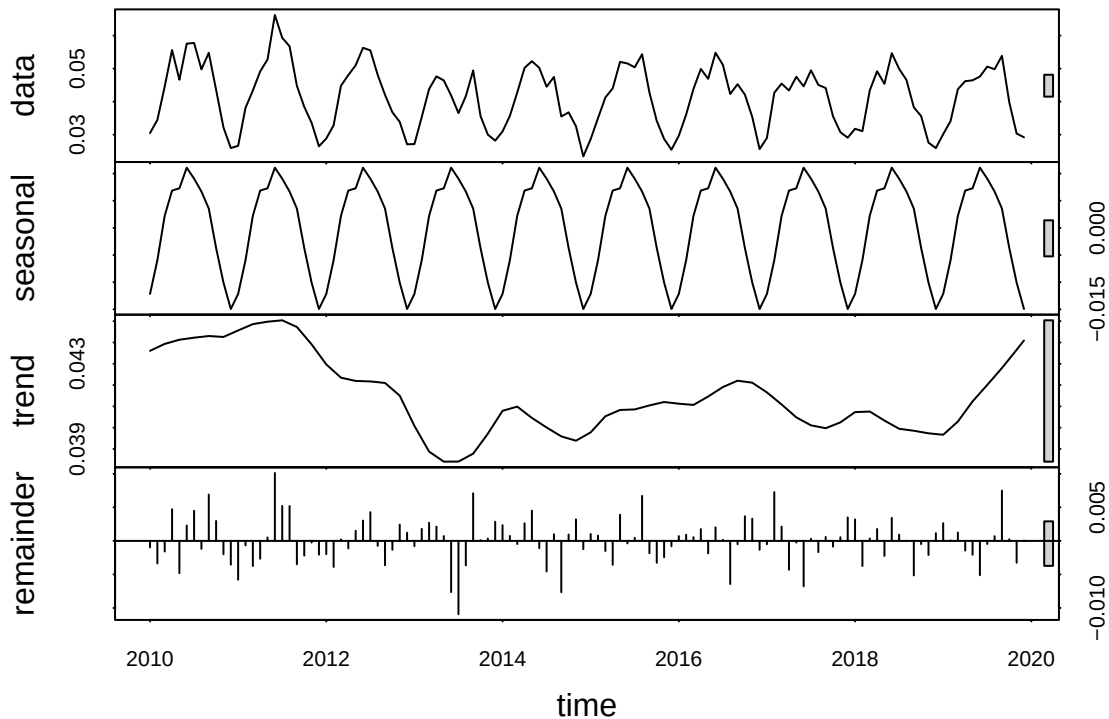
11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
```

```
GaringerOzone.daily.decomposed <- stl(GaringerOzone.daily.ts, s.window = "periodic")  
plot(GaringerOzone.daily.decomposed)
```



```
GaringerOzone.monthly.decomposed <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(GaringerOzone.monthly.decomposed)
```

12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12

#SMK test

Garinger.Ozone.monthly.smk <- SeasonalMannKendall(GaringerOzone.monthly.ts)

Garinger.Ozone.monthly.smk

## tau = -0.143, 2-sided pvalue =0.046724

summary(Garinger.Ozone.monthly.smk)

## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: The seasonal Mann-Kendall is best because there is a clear seasonal pattern in the data.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
```

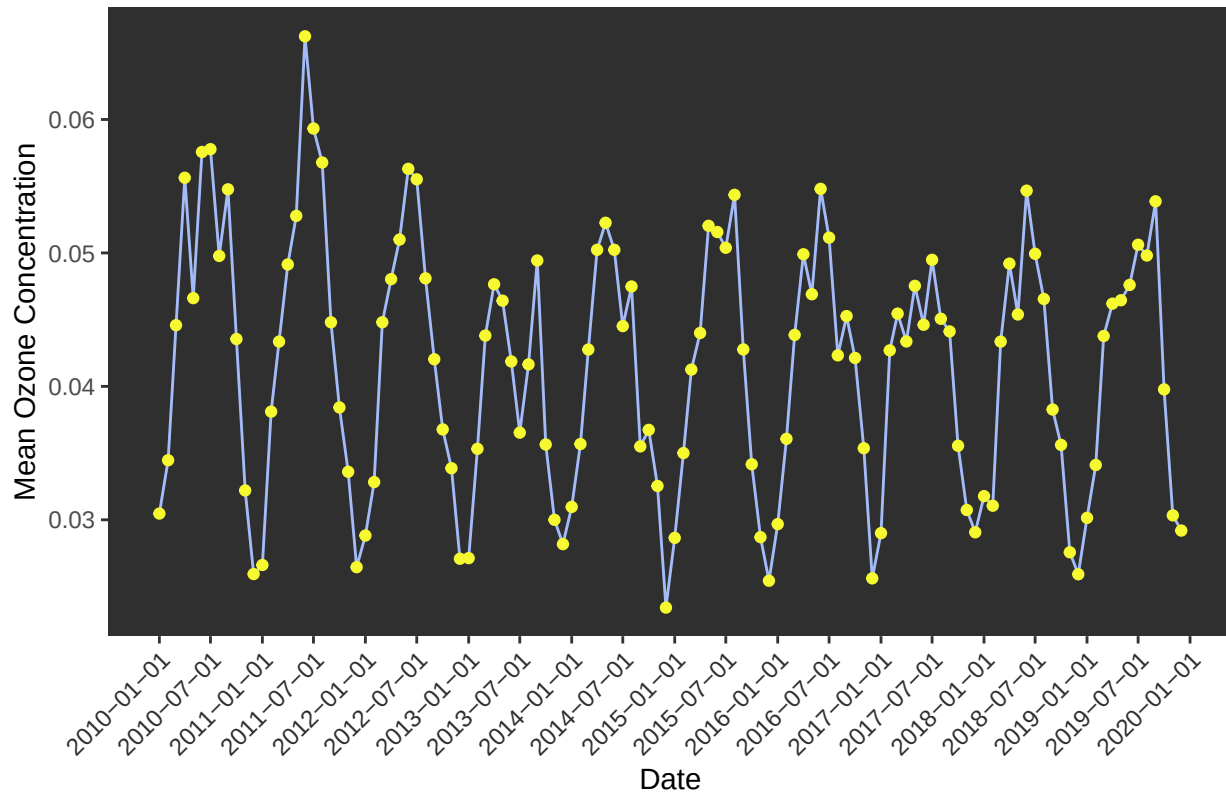
```
monthly.ozone <- Graph.GaringerOzone.monthly %>%
  ggplot(aes(x=graph_date, y=mean_ozone))+
  geom_line(color="#A3BCF9")+
  geom_point(color="#F6F930")+
  labs(
    title="Mean Monthly Ozone Concentrations",
    x="Date",
    y="Mean Ozone Concentration",
    color="#2F2F2F",
  )+
  scale_x_date(
    date_breaks = "6 months"
  )+
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, color="#2F2F2F"),
    panel.grid = element_blank(),
    panel.background = element_rect(fill="#2F2F2F")
  )

print(monthly.ozone)
```

```
## Ignoring unknown labels:
```

```
## * colour : "#2F2F2F"
```

Mean Monthly Ozone Concentrations



```
class(monthly.ozone)
```

```
## [1] "ggplot2::ggplot" "ggplot"          "ggplot2::gg"      "S7_object"
## [5] "gg"
```

14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: The results of the SMK test indicate that there is a statistically significant (pvalue is just under 0.05) but relatively weak negative trend ($\tau = -.143$) in monthly ozone concentration between 2010 and 2019.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
```

```
#subtracting seasonal component.
```

```
GaringerOzone.no.season <- GaringerOzone.monthly.ts - GaringerOzone.monthly.decomposed$time.series[, "s"]
```

#16

```
No.Season.SMK <- MannKendall(GaringerOzone.no.season)
```

```
summary(No.Season.SMK)
```

```
## Score = -1179 , Var(Score) = 194365.7
```

```
## denominator = 7139.5
```

```
## tau = -0.165, 2-sided pvalue =0.0075402
```

Answer: The MK test with no-season data results in a slightly steeper decrease in Ozone over time , $\tau = -0.165$, but the p value is much closer to 0 than the seasonal data and therefore much more significant.